

THREE CONSECUTIVE SUMS OF TWO SQUARES: A POWER SAVING OVER THE SQUARE-ROOT BOUND

ABSTRACT. Let $S(x)$ be the number of $n \leq x$ such that $n-1$, n and $n+1$ are all sums of two squares. One-parameter constructions such as $n = m^2 + 1$ give $S(x) \gg x^{1/2}(\log x)^{-1/2}$. We prove that $S(x) \gg_\delta x^{1/2+\delta}$ for every $\delta < 1/52$.

The integers $n \geq 1$ for which $n^2 - 1$ is a sum of two squares are exactly the integers of the form $n = \frac{1}{2}(d + \frac{b^2+1}{d})$ with $d \geq 1$ and $d \mid b^2+1$, and for them $n \pm 1$ are automatically sums of two squares. For each prime $d \equiv 1 \pmod{8}$ with $d \leq x^\theta$, $\theta < 1/26$, the choice $b = 4w$ turns these into the values of a single quadratic polynomial $N_d(w)$. We evaluate $\sum_w r(N_d(w))$ asymptotically by Hooley's method: Gauss's correspondence between roots of quadratic congruences and representations by binary forms, combined with Weil's bound for Kloosterman sums. All dependence on d is kept polynomial. Summing over d and using $r(n) \ll n^\varepsilon$ to remove multiplicities gives the theorem. No sieve is required.

1. INTRODUCTION

Let $\mathcal{S} = \{a^2 + b^2 : a, b \in \mathbb{Z}\}$ (so $0 \in \mathcal{S}$), and let

$$S(x) = \#\{1 \leq n \leq x : n-1, n, n+1 \in \mathcal{S}\}.$$

Four consecutive integers can never all lie in $\mathcal{S}$, because one of them is $\equiv 3 \pmod{4}$. Triples do occur, for instance $(8, 9, 10)$ and $(16, 17, 18)$.

J. E. Littlewood asked whether, for given unequal positive integers h, k , there are infinitely many n with $n, n+h, n+k \in \mathcal{S}$. Hooley [9] answered this affirmatively using ternary quadratic forms; see also [6, §2.1]. For three consecutive integers infinitude is elementary, for instance $n = (2j^2 + 1)^2$ with $n-1 = (2j^2)^2 + (2j)^2$.

The quantitative question is governed by an analogue, for sums of two squares, of the Hardy–Littlewood prime k -tuple conjecture. It was formulated by Freiberg, Kurlberg and Rosenzweig [6, Conjecture 1.1], following Connors–Keating [2] and Smilansky [14] for pairs. For a finite set $\mathbf{h} \subset \mathbb{Z}$ with non-vanishing singular series $\mathfrak{S}_{\mathbf{h}}$ it predicts

$$\#\{n \leq x : n + \mathbf{h} \subseteq \mathcal{S}\} \sim \mathfrak{S}_{\mathbf{h}} x \left(\frac{C}{\sqrt{\log x}} \right)^{\#\mathbf{h}},$$

where $C = 0.7642\dots$ is the Landau–Ramanujan constant. Freiberg, Kurlberg and Rosenzweig show that this conjecture implies Poisson statistics for the gaps between consecutive sums of two squares, and hence for the level spacings of arithmetic toral point scatterers. For $\mathbf{h} = \{-1, 0, 1\}$ it predicts

$$S(x) \sim \mathfrak{S}_{\{-1,0,1\}} C^3 \frac{x}{(\log x)^{3/2}},$$

and the singular series for this triple was computed by Smilansky [14]. Cochrane and Dressler [1] proved the upper bound $S(x) \ll x(\log x)^{-3/2}$ of this order with the Selberg sieve.

Lower bounds are much weaker. As [6] point out, bounds of the conjectured order are known for pairs, where $\#\{n \leq x : n, n+h \in \mathcal{S}\} \gg x/\log x$ by Hooley [10] and Indlekofer [11], but not for any set $\mathbf{h}$ with $\#\mathbf{h} \geq 3$. Any one-parameter family gives at most about $x^{1/2}$ triples up to x .

For example, with $n = m^2 + 1$ only $m^2 + 2 \in \mathcal{S}$ is needed, and the theorem of Friedlander and Iwaniec [4, Theorem 2] gives $\gg X(\log X)^{-1/2}$ such $m \leq X$, hence

$$S(x) \gg x^{1/2}(\log x)^{-1/2}.$$

As far as we are aware, no bound of the form $S(x) \gg x^{1/2+\delta}$ with $\delta > 0$ has appeared in the literature. Our result is the following. It is a first step beyond the square-root bound towards the case $\mathbf{h} = \{-1, 0, 1\}$ of the conjecture of [6], though still far from the predicted order $x(\log x)^{-3/2}$.

Theorem 1.1. *For every $\delta < 1/52$ we have $S(x) \gg_\delta x^{1/2+\delta}$ for $x \geq 2$.*

For $2 \leq x < x_0$ the theorem holds trivially, because $n = 1$ is always counted ($0, 1, 2 \in \mathcal{S}$). Siegel's theorem is used in Lemma 3.3, so the implied constant is ineffective as stated. Remark 6.1 explains how to make it effective.

1.1. Strategy. The argument carries out the following idea. Suppose $N^2 - 1 = a^2 + b^2$. Then $(N - a)(N + a) = b^2 + 1$, so

$$(1.1) \quad N = C_d(b) := \frac{1}{2} \left(d + \frac{b^2 + 1}{d} \right), \quad d = N - a, \quad d \mid b^2 + 1.$$

Conversely, $N^2 - 1 \in \mathcal{S}$ is equivalent to $N \pm 1 \in \mathcal{S}$ (Lemma 2.1). So the numbers (1.1) automatically have $N \pm 1 \in \mathcal{S}$, and N is the middle of a triple exactly when $C_d(b) \in \mathcal{S}$.

Fix d . As b runs through a residue class of roots of $b^2 \equiv -1 \pmod{d}$, $C_d(b)$ is a quadratic polynomial in the parameter, with about $\sqrt{x/d}$ values up to x . Summing over $d \leq x^\theta$ gives about $x^{1/2+\theta/2}$ values in total. It remains to show that a reasonable proportion of them lie in $\mathcal{S}$, with small multiplicity.

We detect membership in $\mathcal{S}$ with the weight $r(n) = 4 \sum_{k|n} \chi(k)$. This requires the distribution of the values in residue classes modulo k for k up to $\sqrt{x}$. That range exceeds the length $\sqrt{x/d}$ of the polynomial sequence by a factor $\sqrt{d}$. This is the classical situation of Hooley's theorem on $\sum_{n \leq X} \tau(an^2 + bn + c)$ for a quadratic with large leading coefficient [8]. Hooley's method is also the analytic engine of [4]: Lemma 5 there is Gauss's correspondence and Lemma 4 there is Hooley's estimate for incomplete Kloosterman sums. We need this method with explicit polynomial dependence on d (Proposition 4.3). That dependence is affordable once $d \leq x^\theta$ with $\theta < 1/26$.

Since we only aim for a power of x , no sieve is needed. Every element of our family satisfies $r(N) \ll x^\varepsilon$ and has at most x^ε preimages (Proposition 2.5). Restricting to $b = 4w$ and to primes $d \equiv 1 \pmod{8}$ makes $N \equiv 1 \pmod{8}$, which simplifies the bookkeeping.

1.2. Notation. We write $e(t) = e^{2\pi i t}$. The letter χ denotes the non-principal character modulo 4, and

$$r(n) = \#\{(a, b) \in \mathbb{Z}^2 : a^2 + b^2 = n\} = 4 \sum_{k|n} \chi(k) \quad (n \geq 1).$$

For a prime ℓ , v_ℓ is the ℓ -adic valuation. The functions $\tau(n)$ and $\omega(n)$ count the divisors and the distinct prime factors of n . We write $\bar{a}$ for a multiplicative inverse, with the modulus clear from context. For a function g on an interval J , $\|g\|_\infty$ is its supremum and $\text{Var}_J(g)$ its total variation. Throughout, $\varepsilon > 0$ is small, and implied constants may depend on ε (and on the fixed smooth functions F, ω below) but on nothing else unless indicated.

2. THE PARAMETRISATION AND THE REDUCTION

Lemma 2.1. *For an integer $N \geq 1$ we have $N^2 - 1 \in \mathcal{S}$ if and only if $N - 1 \in \mathcal{S}$ and $N + 1 \in \mathcal{S}$.*

Proof. Recall that $n \geq 1$ lies in $\mathcal{S}$ if and only if $v_\ell(n)$ is even for every prime $\ell \equiv 3 \pmod{4}$. For $N = 1$ both statements hold, so let $N \geq 2$. The set $\mathcal{S}$ is closed under multiplication, which gives “ $\Leftarrow$ ”. For “ $\Rightarrow$ ”, note that $\gcd(N-1, N+1) \mid 2$. Hence a prime $\ell \equiv 3 \pmod{4}$ divides at most one of $N \pm 1$, and $v_\ell(N \mp 1) \in \{0, v_\ell(N^2 - 1)\}$. Both values are even. $\square$

Lemma 2.2. *Let $d, u, N \in \mathbb{Z}$ satisfy $2dN = u^2 + d^2 + 1$. Then $N^2 - 1 = (N - d)^2 + u^2$, and if $N \geq 1$ then $N \pm 1 \in \mathcal{S}$. Conversely, if $N^2 - 1 = a^2 + u^2$ then $2dN = u^2 + d^2 + 1$ with $d = N - a$.*

Proof. We have $(N - d)^2 + u^2 = N^2 - 2dN + d^2 + u^2 = N^2 - 1$. The second assertion follows from Lemma 2.1, and the converse is the same identity read backwards. $\square$

In the notation (1.1), $2dC_d(b) = b^2 + d^2 + 1$. We use $b = 4w$ throughout.

Lemma 2.3. *Let $d \equiv 1 \pmod{8}$ and let $w \in \mathbb{Z}$ satisfy $16w^2 + 1 \equiv 0 \pmod{d}$. Then*

$$N_d(w) := C_d(4w) = \frac{16w^2 + d^2 + 1}{2d}$$

is an integer with $N_d(w) \equiv 1 \pmod{8}$ and $N_d(w) \pm 1 \in \mathcal{S}$.

Proof. By hypothesis $d \mid 16w^2 + d^2 + 1$. Also $d^2 \equiv 1 \pmod{16}$, so $16w^2 + d^2 + 1 \equiv 2 \pmod{16}$. Thus $(16w^2 + d^2 + 1)/2$ is an integer $\equiv 1 \pmod{8}$ that is divisible by d . Consequently $N_d(w)$ is an integer with $dN_d(w) \equiv 1 \pmod{8}$, that is, $N_d(w) \equiv 1 \pmod{8}$. Finally, apply Lemma 2.2 with $u = 4w$. $\square$

Lemma 2.4. *For every integer $N \geq 1$,*

$$\#\{(d, w) \in \mathbb{Z}^2 : d \geq 1, w \geq 1, 2dN = 16w^2 + d^2 + 1\} \leq r(N^2 - 1).$$

Proof. By Lemma 2.2, the map $(d, w) \mapsto (N - d, 4w)$ is an injection into the set of representations $N^2 - 1 = a^2 + b^2$. $\square$

Fix once and for all a smooth $F : \mathbb{R} \rightarrow [0, 1]$ supported in $[1/2, 1]$ and not identically zero, and put $c_F := \int_0^1 F(t)t^{-1/2} dt > 0$. For $d \equiv 1 \pmod{8}$ and $x \geq 1$ define

$$(2.1) \quad \Sigma_d(x) := \sum_{\substack{w \geq 1 \\ 16w^2 + 1 \equiv 0 \pmod{d}}} r(N_d(w)) F\left(\frac{N_d(w)}{x}\right).$$

Proposition 2.5. *Let $\mathcal{D}$ be a finite set of positive integers $d \equiv 1 \pmod{8}$, and let $x \geq 2$. Then*

$$\sum_{d \in \mathcal{D}} \Sigma_d(x) \leq S(x) \cdot \max_{N \leq x} r(N) r(N^2 - 1) \ll_\varepsilon S(x) x^\varepsilon.$$

Proof. A term of (2.1) is non-zero only if $N = N_d(w) \leq x$ and $r(N) > 0$, that is, $N \in \mathcal{S}$. In that case N is the middle of a triple by Lemma 2.3. Group the terms according to the value of N . By Lemma 2.4 each N occurs for at most $r(N^2 - 1)$ pairs (d, w) , and $0 \leq F \leq 1$. This gives the first inequality. The second follows from $r(n) \leq 4\tau(n) \ll_\varepsilon n^\varepsilon$. $\square$

Theorem 1.1 therefore follows from the next result, which is proved in §5.

Theorem 2.6. *Let $0 < \theta < 1/26$. There is $\varepsilon_0 = \varepsilon_0(\theta) > 0$ such that for $0 < \varepsilon \leq \varepsilon_0$ and $x \geq x_0(\theta, \varepsilon)$,*

$$\Sigma_d(x) \gg_\varepsilon x^{1/2} d^{-1/2-\varepsilon}$$

for every prime $d \equiv 1 \pmod{8}$ with $d \leq x^\theta$. (The implied constant and x_0 are ineffective because of Siegel’s theorem.)

Deduction of Theorem 1.1. Let $\delta < 1/52$. Choose θ with $2\delta < \theta < 1/26$, and then $\varepsilon \leq \varepsilon_0(\theta)$ so small that $\theta(\frac{1}{2} - \varepsilon) - 2\varepsilon > \delta$. Let $\mathcal{D}$ be the set of primes $d \leq x^\theta$ with $d \equiv 1 \pmod{8}$. The prime number theorem for arithmetic progressions gives

$$\sum_{d \in \mathcal{D}} d^{-1/2-\varepsilon} \geq x^{-\theta(1/2+\varepsilon)} \#\mathcal{D} \gg x^{\theta(1/2-\varepsilon)} (\log x)^{-1}.$$

By Theorem 2.6 and Proposition 2.5,

$$S(x) \gg x^{-\varepsilon} \sum_{d \in \mathcal{D}} \Sigma_d(x) \gg x^{1/2+\theta(1/2-\varepsilon)-\varepsilon} (\log x)^{-1} \gg x^{1/2+\delta}. \quad \square$$

3. ARITHMETIC PRELIMINARIES

3.1. A smoothed hyperbola identity for $r(n)$. Fix a smooth non-increasing function $\lambda_0 : \mathbb{R} \rightarrow [0, 1]$ with $\lambda_0(u) = 1$ for $u \leq -\log 2$ and $\lambda_0(u) = 0$ for $u \geq \log 2$. Put $\lambda(u) = \frac{1}{2}(\lambda_0(u) + 1 - \lambda_0(-u))$ and $\omega(t) = \lambda(\log t)$ for $t > 0$. Then ω is smooth with values in $[0, 1]$, and

$$(3.1) \quad \omega(t) = 1 \ (t \leq \tfrac{1}{2}), \quad \omega(t) = 0 \ (t \geq 2), \quad \omega(t) + \omega(1/t) = 1.$$

Lemma 3.1. *If $N \equiv 1 \pmod{4}$ then $r(N) = 8 \sum_{k|N} \chi(k) \omega(k/\sqrt{N})$.*

Proof. By (3.1), $\sum_{k|N} \chi(k) = \sum_{k|N} \chi(k) \omega(k/\sqrt{N}) + \sum_{k|N} \chi(k) \omega(\sqrt{N}/k)$. In the second sum substitute $k = N/j$ and use $\chi(N/j) = \chi(N)\chi(j) = \chi(j)$, valid because $N \equiv 1 \pmod{4}$ and j is odd. Hence the two sums are equal, and the claim follows from $r(N) = 4 \sum_{k|N} \chi(k)$. $\square$

3.2. Roots of $\nu^2 \equiv -m$. From now on d denotes a prime with $d \equiv 1 \pmod{8}$, and

$$m := d^2 + 1.$$

Since $d^2 \equiv 1 \pmod{8}$ we have $m \equiv 2 \pmod{8}$, and $3 \nmid m$. For $n \geq 1$ let

$$\varrho(n) := \#\{\nu \bmod n : \nu^2 + m \equiv 0 \pmod{n}\}.$$

This function is multiplicative.

Lemma 3.2. (a) *For odd n , $\varrho(n) = \#\{w \bmod n : 16w^2 + m \equiv 0 \pmod{n}\}$.*

(b) *Let ℓ be an odd prime and $j \geq 1$. If $\ell \nmid m$ then $\varrho(\ell^j) = 1 + (\frac{-m}{\ell})$. In all cases $\varrho(\ell^j) \leq 2\ell^{\min(j, v_\ell(m))/2}$. Consequently $\varrho(n) \leq 2^{\omega(n)} \gcd(n, m)^{1/2} \ll_\varepsilon n^\varepsilon d$ for odd n .*

(c) *$\varrho(d^j) = 2$ for $j \geq 1$, and every odd prime divisor of m is $\equiv 1 \pmod{4}$.*

Proof. (a) For odd n the map $w \mapsto 4w$ is a bijection of $\mathbb{Z}/n\mathbb{Z}$.

(b) For $\ell \nmid m$ use Hensel's lemma. Let $a = v_\ell(m) \geq 1$. If $j \leq a$, the congruence becomes $\nu^2 \equiv 0 \pmod{\ell^j}$, which has $\ell^{\lfloor j/2 \rfloor}$ solutions. If $j > a$, a solution must satisfy $v_\ell(\nu) = a/2$, so there is none unless a is even. In that case $\nu = \ell^{a/2} \nu_1$, where ν_1 is determined modulo $\ell^{j-a/2}$ and satisfies $\nu_1^2 \equiv -m\ell^{-a} \pmod{\ell^{j-a}}$. There are at most 2 such ν_1 modulo ℓ^{j-a} , each with $\ell^{a/2}$ lifts modulo $\ell^{j-a/2}$. This proves the bound; the consequence follows by multiplicativity, using $m \leq 2d^2$.

(c) We have $m \equiv 1 \pmod{d}$ and $d \equiv 1 \pmod{4}$, so $\varrho(d^j) = 1 + (\frac{-1}{d}) = 2$. If an odd prime ℓ divides $d^2 + 1$, then -1 is a square modulo ℓ , so $\ell \equiv 1 \pmod{4}$. $\square$

For odd k put $f(k) := \varrho(dk)/2$. By Lemma 3.2, f is multiplicative on odd integers, with

$$(3.2) \quad f(d^j) = 1, \quad f(\ell^j) = \varrho(\ell^j) \quad (\ell \neq d, j \geq 0).$$

Since $\chi(k) = 0$ for even k , the function χf is defined on all of $\mathbb{N}$.

3.3. The singular series $\mathfrak{L}_d$. Define $\psi(n) = \left(\frac{m}{n}\right)$ (Jacobi symbol) for odd $n \geq 1$, and $\psi(n) = 0$ for even n .

Lemma 3.3. (a) ψ is a non-principal real Dirichlet character modulo $4m$.

(b) For $t \geq 1$, $A(t) := \sum_{k \leq t} \chi(k)f(k) \ll_\varepsilon t^{1/2+\varepsilon} d^{1+\varepsilon}$.

(c) The series $\mathfrak{L}_d := \sum_{k \geq 1} \chi(k)f(k)/k$ converges, $|\mathfrak{L}_d| \ll_\varepsilon d^{1+\varepsilon}$, and

$$\mathfrak{L}_d = L(1, \chi) L(1, \psi) H_d \quad \text{with} \quad H_d \geq \frac{6}{\pi^2} \cdot \frac{\phi(dm)}{dm}.$$

(d) $\mathfrak{L}_d \gg_\varepsilon d^{-\varepsilon}$.

Proof. (a) Write $m = 2m'$ with $m' \equiv 1 \pmod{4}$. This is possible because $m \equiv 2 \pmod{8}$. For odd $n > 0$, quadratic reciprocity for Jacobi symbols gives $\left(\frac{m}{n}\right) = \left(\frac{2}{n}\right) \left(\frac{n}{m'}\right)$. This is the product of a character modulo 8 and a character modulo m' , so it is a character modulo $8m' = 4m$ that vanishes on even n .

To see that ψ is non-principal, consider two cases. If m' is not a square, $\left(\frac{\cdot}{m'}\right)$ is non-principal, and the Chinese remainder theorem gives $n \equiv 1 \pmod{8}$ with $\psi(n) = \left(\frac{n}{m'}\right) = -1$. If $m' = s^2$, then $3 \nmid s$ and $\psi(3) = \left(\frac{3}{s}\right)^2 = -1$.

(b) Let $g = \chi * \psi$. We compare Euler factors, writing $X = \ell^{-s}$.

- At $\ell = 2$, both χf and g have Euler factor 1.
- For an odd prime $\ell \nmid dm$, (3.2) and Lemma 3.2(b) give, with $c = \chi(\ell)$ and $\psi(\ell) = c \left(\frac{-m}{\ell}\right)$,

$$\sum_{j \geq 0} \chi(\ell^j) f(\ell^j) X^j = 1 + \left(1 + \left(\frac{-m}{\ell}\right)\right) \frac{cX}{1 - cX} = \frac{1 + \psi(\ell)X}{1 - cX} = \frac{1 - X^2}{(1 - \chi(\ell)X)(1 - \psi(\ell)X)}.$$

- At $\ell = d$, $\chi(d) = 1$ because $d \equiv 1 \pmod{4}$, and $\psi(d) = \left(\frac{m}{d}\right) = 1$ because $m \equiv 1 \pmod{d}$. The factor of χf is $(1 - X)^{-1} = (1 - X) \cdot (1 - X)^{-2}$.
- For odd $\ell \mid m$, $\chi(\ell) = 1$ by Lemma 3.2(c) and $\psi(\ell) = 0$. The factor of χf is $P_\ell(X) := \sum_j \varrho(\ell^j) X^j = (1 - X) P_\ell(X) \cdot (1 - X)^{-1}$.

Hence $\chi f = g * \eta$, where η is multiplicative and determined as follows: $\eta(\ell^j) = 0$ for $\ell = 2$ and $j \geq 1$; $\eta(\ell^2) = -1$ and $\eta(\ell^j) = 0$ for $j \neq 0, 2$ when $\ell \nmid 2dm$; $\eta(d) = -1$ and $\eta(d^j) = 0$ for $j \geq 2$; and $\eta(\ell^j) = \varrho(\ell^j) - \varrho(\ell^{j-1})$ for odd $\ell \mid m$. By Lemma 3.2(b), for odd $\ell \mid m$ with $a = v_\ell(m)$,

$$\sum_{j \geq 0} |\eta(\ell^j)| \ell^{-j(1/2+\varepsilon)} \leq 1 + 2a + 2 \sum_{i \geq 1} \ell^{-i/2} \leq 2a + 5.$$

Therefore $\sum_{n \geq 1} |\eta(n)| n^{-1/2-\varepsilon} \leq \zeta(1+2\varepsilon) (1 + d^{-1/2}) \prod_{\ell \mid m, \ell \text{ odd}} (2v_\ell(m) + 5) \ll_\varepsilon d^\varepsilon$.

Next, let $\Psi^* = \sup_y |\sum_{n \leq y} \psi(n)|$. Applying the Pólya–Vinogradov inequality [3, Ch. 23] to the primitive character inducing ψ , and Möbius inversion over the remaining primes dividing $4m$, gives $\Psi^* \ll_\varepsilon m^{1/2+\varepsilon}$. The hyperbola method with parameter $\sqrt{y}$ gives, for $y \geq 1$,

$$\left| \sum_{n \leq y} g(n) \right| = \left| \sum_{a \leq \sqrt{y}} \chi(a) \sum_{b \leq y/a} \psi(b) + \sum_{b \leq \sqrt{y}} \psi(b) \sum_{a \leq y/b} \chi(a) - \sum_{a \leq \sqrt{y}} \chi(a) \sum_{b \leq \sqrt{y}} \psi(b) \right| \leq 3\sqrt{y} (\Psi^* + 1).$$

Finally, $A(t) = \sum_{e \leq t} \eta(e) \sum_{n \leq t/e} g(n)$, so

$$|A(t)| \ll_\varepsilon m^{1/2+\varepsilon} \sum_{e \leq t} |\eta(e)| (t/e)^{1/2} \leq m^{1/2+\varepsilon} t^{1/2+\varepsilon} \sum_e |\eta(e)| e^{-1/2-\varepsilon} \ll_\varepsilon t^{1/2+\varepsilon} d^{1+3\varepsilon}.$$

Replacing ε by $\varepsilon/3$ gives (b).

(c) Convergence of the series and the bound $|\mathfrak{L}_d| \leq \int_1^\infty |A(u)|u^{-2} du \ll d^{1+\varepsilon}$ follow from (b) by partial summation. For $\Re s > 1$ we have

$$\sum_k \chi(k)f(k)k^{-s} = L(s, \chi)L(s, \psi)H_d(s), \quad H_d(s) = \sum_n \eta(n)n^{-s}.$$

By (b), the left-hand side is a Dirichlet series converging, and hence analytic, for $\Re s > 1/2$. The right-hand side is analytic there because χ and ψ are non-principal and $H_d(s)$ converges absolutely. So the identity persists to $s = 1$, with $H_d := H_d(1)$. Its Euler factors are $1 - \ell^{-2}$ for $\ell \nmid 2dm$, $1 - d^{-1}$ at d , $(1 - \ell^{-1})P_\ell(\ell^{-1}) \geq 1 - \ell^{-1}$ for odd $\ell \mid m$, and 1 at 2. This gives the stated lower bound.

(d) We have $L(1, \chi) = \pi/4$. Let ψ^* be the primitive character of conductor $q^* \mid 4m$ inducing ψ . Then $L(1, \psi) = L(1, \psi^*) \prod_{\ell \mid 4m} (1 - \psi^*(\ell)/\ell) \geq L(1, \psi^*)\phi(4m)/(4m)$. Siegel's theorem [3, Ch. 21] gives $L(1, \psi^*) \gg_\varepsilon (q^*)^{-\varepsilon}$. Combining these with (c) and $\phi(n)/n \gg (\log \log 3n)^{-1}$ proves (d). $\square$

4. WEYL SUMS FOR ROOTS OF $16w^2 + m$, UNIFORMLY IN d

4.1. Gauss's correspondence. We use binary forms $\varphi = [A, 2B, C]$, meaning $\varphi(X, Y) = AX^2 + 2BXY + CY^2$ with $A, B, C \in \mathbb{Z}$, $A > 0$ and determinant $AC - B^2 = \Delta > 0$. The group $SL_2(\mathbb{Z})$ acts by $\varphi \circ \sigma(X, Y) = \varphi(RX + TY, SX + VY)$ for $\sigma = \begin{pmatrix} R & T \\ S & V \end{pmatrix}$.

The next lemma is classical; compare [4, Lemma 5 and its Corollary]. We include the short proof because we need the exact form of the correspondence.

Lemma 4.1. *Let $\Delta \geq 1$ be an integer such that neither Δ nor $\Delta/3$ is a perfect square. Let $\mathcal{R}_\Delta$ be a set of representatives of the $SL_2(\mathbb{Z})$ -classes of positive definite forms $[A, 2B, C]$ of determinant Δ , each chosen reduced, i.e. $|2B| \leq A \leq C$. Let $\mathcal{M} \geq 1$. Given a form $\varphi = [A, 2B, C] \in \mathcal{R}_\Delta$ and a primitive vector $(R, S) \in \mathbb{Z}^2$ with $\varphi(R, S) = \mathcal{M}$, complete it to $\sigma = \begin{pmatrix} R & T \\ S & V \end{pmatrix} \in SL_2(\mathbb{Z})$ and set*

$$\omega := T(AR + BS) + V(BR + CS).$$

This defines a bijection

$$\{(\varphi, \pm(R, S))\} \longrightarrow \{\omega \bmod \mathcal{M} : \omega^2 + \Delta \equiv 0 \pmod{\mathcal{M}}\}.$$

Moreover, if $S \neq 0$ then

$$(4.1) \quad \frac{\omega}{\mathcal{M}} \equiv \frac{\bar{R}}{S} - \frac{AR + BS}{S\mathcal{M}} \pmod{1}, \quad \bar{R}R \equiv 1 \pmod{|S|}.$$

Proof. We have $\varphi \circ \sigma = [\mathcal{M}, 2\omega, C']$ with $\mathcal{M}C' - \omega^2 = \Delta$, so $\omega^2 + \Delta \equiv 0 \pmod{\mathcal{M}}$. Any other completion has the form $\sigma \begin{pmatrix} 1 & j \\ 0 & 1 \end{pmatrix}$, which replaces ω by $\omega + j\mathcal{M}$. Replacing σ by $-\sigma$ does not change ω . So the map is well defined.

Surjectivity: if $\omega^2 + \Delta \equiv 0 \pmod{\mathcal{M}}$, the form $\Phi = [\mathcal{M}, 2\omega, (\omega^2 + \Delta)/\mathcal{M}]$ is positive definite of determinant Δ . Hence $\Phi = \varphi \circ \sigma$ for some $\varphi \in \mathcal{R}_\Delta$ and $\sigma \in SL_2(\mathbb{Z})$, and the first column of σ is a preimage of ω .

Injectivity: two preimages of the same ω yield, after adjusting the completions, $\varphi \circ \sigma = \varphi' \circ \sigma'$. Hence $\varphi = \varphi'$, and $\sigma'\sigma^{-1}$ is a proper automorph of φ . A positive definite form with a proper automorph other than $\pm I$ is equivalent to a multiple of $X^2 + Y^2$ or of $X^2 + XY + Y^2$. Among forms with even middle coefficient these multiples are $[c, 0, c]$ and $[2g, 2g, 2g]$, of determinants c^2 and $3g^2$, which the hypothesis on Δ excludes. Thus $\sigma' = \pm\sigma$.

Finally, put $P = AR + BS$ and $Q = BR + CS$. Then $RP + SQ = \varphi(R, S) = \mathcal{M}$ and $TP + VQ = \omega$. Since $RV - ST = 1$, solving gives $P = V\mathcal{M} - S\omega$. Hence $\omega/\mathcal{M} = V/S - P/(S\mathcal{M})$ when $S \neq 0$, and $RV \equiv 1 \pmod{|S|}$. This is (4.1). $\square$

4.2. Incomplete Kloosterman sums in progressions.

Lemma 4.2. *Let $L \geq 2$ be even, $S \geq 1$, $R_0, a \in \mathbb{Z}$, and let $J \subset \mathbb{R}$ be an interval of length at most ℓ . Write $S = S_1 S_2$, where every prime factor of S_1 divides L and $(S_2, L) = 1$. Then*

$$\left| \sum_{\substack{R \in J, \\ (R, S) = 1}} e\left(\frac{a\bar{R}}{S}\right) \right| \leq \tau(S_2)(a, S_2)^{1/2} \left(\frac{\ell}{L S_2^{1/2}} + 3S_1 S_2^{1/2} \log(2LS) \right).$$

Proof. Let $q = \text{lcm}(L, S) = q_1 S_2$ with $q_1 = \text{lcm}(L, S_1)$, and let $\phi(\xi)$ be the summand. It is periodic modulo q . Completing the sum gives

$$\left| \sum_{R \in J} \phi(R) \right| \leq \max_{\eta \bmod q} |\mathcal{K}_\eta| \cdot \frac{\ell + 3q \log(2q)}{q}, \quad \mathcal{K}_\eta = \sum_{\xi \bmod q} \phi(\xi) e(-\eta \xi / q),$$

using $\sum_{\eta \bmod q} \min(\ell + 1, \|\eta/q\|^{-1}/2) \leq \ell + 3q \log(2q)$.

Since $(q_1, S_2) = 1$, we have $\frac{\bar{\xi}}{S} \equiv \frac{\bar{\xi} \bar{S}_2}{S_1} + \frac{\bar{\xi} \bar{S}_1}{S_2} \pmod{1}$, and $e(-\eta \xi / q)$ splits similarly. By the Chinese remainder theorem, $\mathcal{K}_\eta$ is therefore the product of a sum over $\xi_1 \bmod q_1$ with $\xi_1 \equiv R_0 \pmod{L}$ and a Kloosterman sum $S(a \bar{S}_1, \eta'; S_2)$. We bound the first factor trivially by $q_1 / L \leq S_1$. For the second, Weil's bound $|S(u, v; c)| \leq \tau(c)(u, v, c)^{1/2} c^{1/2}$ [12, Cor. 11.12] gives at most $\tau(S_2)(a, S_2)^{1/2} S_2^{1/2}$. Inserting $|\mathcal{K}_\eta| \leq (q_1 / L) \tau(S_2)(a, S_2)^{1/2} S_2^{1/2}$ above, and using $q \leq LS$, proves the lemma. $\square$

4.3. The main estimate. For odd $\mu \geq 1$ and $h \in \mathbb{Z}$ put

$$S_\mu(h) := \sum_{\substack{w \bmod \mu \\ 16w^2 + m \equiv 0 \pmod{\mu}}} e\left(\frac{hw}{\mu}\right),$$

so that $S_\mu(0) = \varrho(\mu)$ by Lemma 3.2(a). The following proposition is the uniform version of Hooley's estimate that we need. It holds for any odd $d \geq 1$ with $m = d^2 + 1$.

Proposition 4.3. *Let $d \geq 1$ be odd, $m = d^2 + 1$ and $c \in \{1, 3\}$. Let $M \geq m$, $h \in \mathbb{Z} \setminus \{0\}$, and let $\beta : (M, 2M] \rightarrow \mathbb{C}$ satisfy $\|\beta\|_\infty + \text{Var}(\beta) \leq B_0$. Then*

$$\Xi := \sum_{\substack{M < \mu \leq 2M \\ d|\mu, \mu/d \equiv c \pmod{4}}} \beta(\mu) S_\mu(h) \ll_\varepsilon B_0 |h|^{3/2} d m^{1/8} M^{3/4} (|h| d M)^\varepsilon.$$

Proof. Every μ in the sum is odd. Put $\Delta = 64m$. Neither Δ nor $\Delta/3$ is a square, since m is not a square and $3 \nmid m$.

Step 1: a reformulation. Let ω run over the residues modulo 64μ with $\omega^2 + \Delta \equiv 0 \pmod{64\mu}$ and $32 \mid \omega$. Writing $\omega = 32w'$ with w' modulo 2μ , the condition becomes $16w'^2 + m \equiv 0 \pmod{\mu}$, and $e(2h\omega/(64\mu)) = e(hw'/\mu)$. The reduction $w' \bmod 2\mu \mapsto w' \bmod \mu$ is 2-to-1, so

$$S_\mu(h) = \frac{1}{2} \sum_{\substack{\omega \bmod 64\mu \\ \omega^2 + \Delta \equiv 0 \pmod{64\mu}, 32 \mid \omega}} e\left(\frac{2h\omega}{64\mu}\right).$$

Step 2: passage to forms. Apply Lemma 4.1 with $\mathcal{M} = 64\mu$. The condition $32 \mid \omega$ makes sense because $32 \mid \mathcal{M}$. A reduced form satisfies $A \leq \sqrt{4\Delta/3} < 10\sqrt{m} < 64\mu$, so $S = 0$ cannot occur, and we may take $S \geq 1$. By (4.1),

$$\Xi = \frac{1}{2} \sum_{\varphi \in \mathcal{R}_\Delta} \sum_{S \geq 1} \sum_{R \in \mathcal{I}_{\varphi, S}}^\dagger \beta\left(\frac{\varphi(R, S)}{64}\right) e\left(-\frac{2h(AR + BS)}{S \varphi(R, S)}\right) e\left(\frac{2h\bar{R}}{S}\right),$$

where $\mathcal{I}_{\varphi,S} = \{R \in \mathbb{Z} : 64M < \varphi(R, S) \leq 128M\}$. The dagger restricts to $(R, S) = 1$ and to the conditions

$$(4.2) \quad (i) \ 64d \mid \varphi(R, S), \quad (ii) \ \varphi(R, S)/(64d) \equiv c \pmod{4}, \quad (iii) \ 32 \mid \omega(\varphi; R, S).$$

Step 3: the conditions are periodic. Put $L = 256d$. Conditions (i) and (ii) depend only on $(R, S) \bmod L$. For (iii), suppose $(R', S') \equiv (R, S) \pmod{32}$, with completions (T, V) and (T', V') . Then $R(V' - V) - S(T' - T) \equiv 0 \pmod{32}$. Writing $(T' - T, V' - V) = \alpha(R, S) + \beta'(T, V)$, this says $\beta' \equiv 0 \pmod{32}$. Hence $\omega' \equiv \omega + \alpha\varphi(R, S) \equiv \omega \pmod{32}$ whenever (i) holds. Consequently, for fixed S , whether R satisfies (i)–(iii) depends only on $R \bmod L$, so the admissible R are the union of certain residue classes $R_0 \bmod L$, at most L of them.

Step 4: fixed φ and S . Since $A\varphi(R, S) = (AR + BS)^2 + \Delta S^2$, the set $\mathcal{I}_{\varphi,S}$ is empty unless

$$S \leq S_{\varphi} := \sqrt{128AM/\Delta} = \sqrt{2AM/m}.$$

Moreover, it consists of the integers in at most two intervals, each of length at most $\ell_{\varphi} := 2\sqrt{128M/A}$, on each of which $R \mapsto \varphi(R, S)$ is monotone. On such an interval J let $g(R) = \beta(\varphi(R, S)/64) e(-2h(AR + BS)/(S\varphi(R, S)))$. Using $(AR + BS)^2 \leq A\varphi(R, S)$,

$$\left| \frac{d}{dR} \frac{2h(AR + BS)}{S\varphi(R, S)} \right| \leq \frac{6|h|A}{S\varphi(R, S)} \leq \frac{6|h|A}{64MS},$$

so the phase varies by at most $\frac{6|h|A}{64M}\ell_{\varphi} \leq 3|h|\sqrt{A/M} \leq 10|h|$ on J , because $A/M < 10\sqrt{m}/m \leq 10$. Hence $\|g\|_{\infty} + \text{Var}_J(g) \leq B_0(2 + 20\pi|h|) \leq 70B_0|h|$.

By partial summation and Lemma 4.2 (with $a = 2h$, and noting $(2h, S_2) = (h, S_2) \leq |h|$), the contribution of (φ, S) is

$$\ll B_0|h|^{3/2} \tau(S) \left(\frac{\ell_{\varphi}}{S_2^{1/2}} + L S_1 S_2^{1/2} \log(2LS) \right),$$

after summing over the at most L admissible classes $R_0 \bmod L$ and the at most two intervals.

Step 5: summation over S and φ . Group the S by their L -part S_1 . We have $\sum_{s_1} s_1^{-1/2} = \prod_{\ell|L} (1 - \ell^{-1/2})^{-1} \ll_{\varepsilon} L^{\varepsilon}$, where s_1 runs over integers composed of primes dividing L . Hence

$$\sum_{S \leq X} S_2^{-1/2} \ll_{\varepsilon} X^{1/2} L^{\varepsilon}, \quad \sum_{S \leq X} S_1 S_2^{1/2} \ll_{\varepsilon} X^{3/2} L^{\varepsilon}.$$

Also $S_{\varphi} \leq \sqrt{20M}$, so $\tau(S) \ll M^{\varepsilon}$. The contribution of a fixed φ is therefore

$$\ll B_0|h|^{3/2} (LM)^{2\varepsilon} (\ell_{\varphi} S_{\varphi}^{1/2} + L S_{\varphi}^{3/2}) \ll B_0|h|^{3/2} (LM)^{2\varepsilon} M^{3/4} (A^{-1/4} m^{-1/4} + L A^{3/4} m^{-3/4}).$$

A form in $\mathcal{R}_{\Delta}$ with first coefficient A is determined by B , which satisfies $|2B| \leq A$ and $B^2 + \Delta \equiv 0 \pmod{A}$. So there are at most $2\varrho_{\Delta}(A)$ such forms, where $\varrho_{\Delta}(A) = \#\{B \bmod A : B^2 + \Delta \equiv 0 \pmod{A}\}$. As in Lemma 3.2(b), $\varrho_{\Delta}(\ell^j) \leq 2\ell^{\min(j, v_{\ell}(m))/2}$ for odd ℓ . Since $\Delta = 2^7 m/2$ with $m/2$ odd, $\varrho_{\Delta}(2^j) \leq 8$, with $\varrho_{\Delta}(2^j) = 0$ for $j \geq 8$. Hence $\varrho_{\Delta}(A) \leq 8 \cdot 2^{\omega(A)} \gcd(A, m)^{1/2}$, and

$$\sum_{A \leq Y} \varrho_{\Delta}(A) \leq 8 \sum_{e|m} \sqrt{e} 2^{\omega(e)} \sum_{A' \leq Y/e} 2^{\omega(A')} \ll_{\varepsilon} Y \log(2Y) m^{\varepsilon}.$$

By partial summation over $A < 10\sqrt{m}$,

$$\sum_{\varphi \in \mathcal{R}_{\Delta}} (A^{-1/4} m^{-1/4} + L A^{3/4} m^{-3/4}) \ll_{\varepsilon} m^{\varepsilon} (m^{3/8-1/4} + L m^{7/8-3/4}) \ll L m^{1/8+\varepsilon}.$$

Altogether $\Xi \ll B_0|h|^{3/2} L m^{1/8} M^{3/4} (LMm)^{3\varepsilon}$. Since $L = 256d$ and $m \leq 2d^2$, renaming ε gives the proposition. $\square$

Remark 4.4. The trivial bound is $\Xi \ll B_0 \sum_{\mu} \varrho(\mu) \ll B_0 M^{1+\varepsilon}/d$, the sum being over $\mu \in (M, 2M]$ with $d \mid \mu$. Proposition 4.3 saves a power of M at a cost polynomial in d and h , which is all we need. Numerically the sums show square-root cancellation in M (see §7).

5. THE ASYMPTOTIC FORMULA FOR $\Sigma_d(x)$

Throughout this section $d \equiv 1 \pmod{8}$ is prime, $m = d^2 + 1$, $0 < \varepsilon < 1/100$, and

$$(5.1) \quad d \leq x^{1/20}, \quad x \geq x_0(\varepsilon).$$

Proposition 5.1. *Under (5.1),*

$$\Sigma_d(x) = 2\sqrt{2} c_F \mathfrak{L}_d \sqrt{x/d} + O_\varepsilon(x^{1/4+\varepsilon} d^{1/2+\varepsilon} + x^{3/8+\varepsilon} d^{11/4}).$$

5.1. Set-up. For real w put $N(w) = (16w^2 + m)/(2d)$, and let $0 < w_- < w_+$ be defined by $N(w_-) = x/2$ and $N(w_+) = x$. Explicitly $w_- = \frac{1}{4}\sqrt{dx - m}$ and $w_+ = \frac{1}{4}\sqrt{2dx - m}$. Under (5.1) we have $\frac{1}{6}\sqrt{dx} \leq w_- < w_+ \leq \frac{1}{2}\sqrt{dx}$. For real $\kappa > 0$ define

$$\Phi_\kappa(w) = F(N(w)/x) \omega(\kappa/\sqrt{N(w)}) \quad (w > 0), \quad \Phi_\kappa(w) = 0 \quad (w \leq 0).$$

Then Φ_κ is smooth, takes values in $[0, 1]$, is supported in $[w_-, w_+]$, and vanishes identically if $\kappa \geq 2\sqrt{x}$.

Lemma 5.2. *Uniformly in $\kappa > 0$ we have $\Phi_\kappa^{(j)}(w) \ll_j (dx)^{-j/2}$ for $j \geq 0$. Consequently, with $\hat{\Phi}_\kappa(\xi) = \int \Phi_\kappa(w) e(-\xi w) dw$,*

$$|\hat{\Phi}_\kappa(\xi)| \leq \frac{1}{2}\sqrt{dx} \quad \text{and} \quad \hat{\Phi}_\kappa(\xi) \ll_j \sqrt{dx} (|\xi|\sqrt{dx})^{-j} \quad (\xi \neq 0).$$

Proof. On $[w_-, w_+]$ we have $N \asymp x$, $N'(w) = 16w/d \asymp \sqrt{x/d}$, $N'' = 16/d$, and $N''' = 0$. Hence $\frac{d^i}{dw^i}(N/x) \ll (dx)^{-i/2}$ and $\frac{d^i}{dw^i} N^{-1/2} \ll x^{-1/2} (dx)^{-i/2}$ for $i \geq 1$. The derivatives $\omega^{(l)}(\kappa N^{-1/2})$ with $l \geq 1$ vanish unless $\kappa N^{-1/2} \in [\frac{1}{2}, 2]$, in which case $\kappa \asymp \sqrt{x}$. The first claim now follows from the Faà di Bruno formula. The others follow by integrating by parts j times over an interval of length at most $\frac{1}{2}\sqrt{dx}$. $\square$

Let $w \geq 1$ with $d \mid 16w^2 + 1$. Then $N(w) \equiv 1 \pmod{4}$ by Lemma 2.3, and Lemma 3.1 applies. For odd k we have $k \mid N(w)$ if and only if $dk \mid 16w^2 + m$, because $2dN(w) = 16w^2 + m$ is even and dk is odd. Also $d \mid 16w^2 + m$ if and only if $d \mid 16w^2 + 1$. Therefore

$$(5.2) \quad \Sigma_d(x) = 8 \sum_{k \text{ odd}} \chi(k) \sum_{\substack{w \in \mathbb{Z} \\ 16w^2 + m \equiv 0 \pmod{dk}}} \Phi_k(w),$$

where only $k < 2\sqrt{x}$ contribute. Poisson summation in each residue class modulo dk gives

$$\sum_{\substack{w \in \mathbb{Z} \\ 16w^2 + m \equiv 0 \pmod{dk}}} \Phi_k(w) = \frac{\varrho(dk)}{dk} \hat{\Phi}_k(0) + \frac{1}{dk} \sum_{h \neq 0} \hat{\Phi}_k\left(\frac{h}{dk}\right) S_{dk}(h).$$

Accordingly we write $\Sigma_d(x) = \mathcal{M}_d + \mathcal{E}_d$, where $\mathcal{M}_d$ collects the terms $h = 0$.

5.2. The main term.

Lemma 5.3. $\mathcal{M}_d = 2\sqrt{2} c_F \mathfrak{L}_d \sqrt{x/d} + O_\varepsilon(x^{1/4+\varepsilon} d^{1/2+\varepsilon})$.

Proof. Since $\varrho(dk) = 2f(k)$ and the k -sum is finite,

$$\mathcal{M}_d = \frac{16}{d} \int_0^\infty F(N(w)/x) G(\sqrt{N(w)}) dw, \quad G(y) := \sum_{k \geq 1} \frac{\chi(k) f(k)}{k} \omega(k/y).$$

By (3.1), $\mathfrak{L}_d - G(y) = \sum_k \frac{\chi(k)f(k)}{k} (1 - \omega(k/y))$ involves only $k \geq y/2$. Partial summation with Lemma 3.3(b) gives

$$\mathfrak{L}_d - G(y) = - \int_{y/2}^{\infty} A(t) \frac{d}{dt} \left(\frac{1 - \omega(t/y)}{t} \right) dt \ll_{\varepsilon} y^{-1/2+\varepsilon} d^{1+\varepsilon}.$$

Now substitute $N = N(w)$, so that $dw = dN/(4\sqrt{2dN-m})$. This gives

$$\mathcal{M}_d = 4 \int_{x/2}^x \frac{F(N/x) G(\sqrt{N})}{\sqrt{2dN-m}} dN = 4 \int \frac{F(N/x)}{\sqrt{2dN}} dN \cdot \mathfrak{L}_d + O_{\varepsilon} \left(\sqrt{\frac{x}{d}} x^{-1/4+\varepsilon} d^{1+\varepsilon} + \sqrt{\frac{x}{d}} \frac{m}{dx} d^{1+\varepsilon} \right),$$

using $|\mathfrak{L}_d| \ll d^{1+\varepsilon}$. Finally $4 \int F(N/x) (2dN)^{-1/2} dN = 2\sqrt{2} c_F \sqrt{x/d}$, and the last error term is negligible. $\square$

5.3. The error term.

Lemma 5.4. $\mathcal{E}_d \ll_{\varepsilon} x^{3/8+\varepsilon} d^{11/4}$.

Proof. Let $K_0 = \sqrt{x/d} x^{-\varepsilon}$.

Small k . For $k \leq K_0$ and $h \neq 0$ we have $|h|\sqrt{dx}/(dk) \geq x^{\varepsilon}|h|$. Lemma 5.2 with $j = \lceil 3/\varepsilon \rceil$, together with $|S_{dk}(h)| \leq \varrho(dk) \ll x$, shows that these terms contribute $O(x^{-1})$ in total.

Large k . Split $K_0 < k < 2\sqrt{x}$ into $O(\log x)$ ranges $k \in I_K \subseteq (K, 2K]$ with $K_0/2 \leq K < 2\sqrt{x}$, and put $H_K = 2K\sqrt{d/x} x^{\varepsilon}$. For $k \in I_K$ and $|h| > H_K$ we have $|h|\sqrt{dx}/(dk) \geq x^{\varepsilon}|h|/H_K$, so these terms also contribute $O(x^{-1})$. Writing $\mu = dk$ and splitting k according to $k \equiv c \pmod{4}$, we obtain

$$\mathcal{E}_d = 8 \sum_K \sum_{1 \leq |h| \leq H_K} \sum_{c \in \{1,3\}} \chi(c) \sum_{\substack{dK < \mu \leq 2dK \\ d|\mu, \mu/d \equiv c \pmod{4}}} \beta_{h,K}(\mu) S_{\mu}(h) + O(x^{-1}),$$

where $\beta_{h,K}(\mu) = \mu^{-1} \hat{\Phi}_{\mu/d}(h/\mu) \mathbf{1}_{I_K}(\mu/d)$.

Bounds for the weights. By Lemma 5.2, $\|\beta_{h,K}\|_{\infty} \leq \sqrt{dx}/(dK)$. Differentiating under the integral sign,

$$\partial_{\mu} \beta_{h,K} = -\frac{\beta_{h,K}}{\mu} + \frac{1}{\mu} \int F \omega' \left(\frac{\mu}{d\sqrt{N(w)}} \right) \frac{e(-hw/\mu)}{d\sqrt{N(w)}} dw + \frac{1}{\mu} \int \Phi_{\mu/d}(w) \frac{2\pi i h w}{\mu^2} e(-hw/\mu) dw$$

on the interior of I_K . Integrating over $\mu \in (dK, 2dK]$, and using $K < 2\sqrt{x}$ and $|h| \leq H_K$, the three terms contribute

$$\ll \frac{\sqrt{dx}}{dK}, \quad \ll \frac{1}{\sqrt{d}} \ll \frac{\sqrt{dx}}{dK}, \quad \ll \frac{|h|x}{dK^2} \ll x^{\varepsilon} \frac{\sqrt{dx}}{dK}.$$

With the jumps of the indicator, $\|\beta_{h,K}\|_{\infty} + \text{Var}(\beta_{h,K}) \ll x^{\varepsilon} \sqrt{dx}/(dK)$.

Application of Proposition 4.3. Take $M = dK$. This is admissible because $dK \geq \frac{1}{2} \sqrt{dx} x^{-\varepsilon} \geq m$ by (5.1). The proposition gives

$$\mathcal{E}_d \ll x^{2\varepsilon} \sum_K \frac{\sqrt{dx}}{dK} d m^{1/8} (dK)^{3/4} \sum_{1 \leq |h| \leq H_K} |h|^{3/2} \ll x^{2\varepsilon} \sum_K \sqrt{dx} d K^{-1/4} H_K^{5/2},$$

using $m^{1/8} \leq 2d^{1/4}$. Since $H_K^{5/2} \ll K^{5/2} d^{5/4} x^{-5/4} x^{3\varepsilon}$, the summand is $\ll x^{5\varepsilon} d^{11/4} x^{-3/4} K^{9/4}$. Summing over dyadic $K < 2\sqrt{x}$ gives $\ll x^{3/8+5\varepsilon} d^{11/4}$. Renaming ε proves the lemma. $\square$

Proposition 5.1 follows from Lemmas 5.3 and 5.4.

Proof of Theorem 2.6. Let $\theta < 1/26$ and $d \leq x^\theta$. By Lemma 3.3(d), the main term in Proposition 5.1 is $\geq c(\varepsilon) x^{1/2} d^{-1/2-\varepsilon}$. The ratio of the error terms to this quantity is

$$\ll_\varepsilon x^{-1/4+\varepsilon} d^{1+2\varepsilon} + x^{-1/8+\varepsilon} d^{13/4+\varepsilon} \leq x^{-1/8+\varepsilon+\theta(13/4+\varepsilon)} + x^{-1/4+\varepsilon+\theta(1+2\varepsilon)}.$$

Since $13\theta/4 < 1/8$, both exponents are negative once ε is small enough in terms of θ . Hence $\Sigma_d(x) \geq \frac{1}{2}c(\varepsilon)x^{1/2}d^{-1/2-\varepsilon}$ for $x \geq x_0(\theta, \varepsilon)$. $\square$

This completes the proof of Theorem 1.1.

6. REMARKS

Remark 6.1 (Effectivity). Siegel's theorem enters only through Lemma 3.3(d). It can be avoided as follows. By the Landau–Page theorem, among real primitive characters of conductor at most $4x^{2\theta} + 4$ there is at most one, ψ_1 , with $L(1, \psi_1) < c/\log x$. The primitive character attached to d depends only on the squarefree part s of $d^2 + 1$ (so $d^2 + 1 = st^2$ with s squarefree). For fixed s , the equation $d^2 - sy^2 = -1$ has $O(\log x)$ solutions with $d \leq x$, because its solutions grow geometrically. Hence ψ_1 is attached to $O(\log x)$ primes d , which we discard. For the remaining primes, $\mathfrak{L}_d \gg (\log x)^{-1}(\log \log x)^{-2}$ effectively, and the rest of the argument is unchanged.

Remark 6.2 (The exponent). The value $1/52 = \frac{1}{2} \cdot \frac{1}{26}$ is not optimal, and we have made no attempt to optimise it. Three improvements are available.

- In Step 3 of Proposition 4.3, for $S \not\equiv 0 \pmod{d}$ only $O(1)$ residue classes $R_0 \pmod{L}$ are admissible. Exploiting this replaces the factor L by $O(\sqrt{L})$ and gives any $\delta < 1/44$.
- One can use cancellation in the h -sum or between different d . Here the averaging over d is used only to add up positive main terms.
- One can use spectral bounds for Weyl sums of quadratic roots that are uniform in the discriminant. See for example Grimmelt–Merikoski [7] for uniform Type I estimates for the congruence $al^2 + h \equiv 0 \pmod{k}$.

The conjectured order $x(\log x)^{-3/2}$ remains far out of reach.

Remark 6.3 (Relation to the Friedlander–Iwaniec method). §4 is a non-bilinear analogue of the Proposition in [4, §4]. Their Lemma 5 and its Corollary are our Lemma 4.1; their Lemma 4, Hooley's bound for incomplete Kloosterman sums, is replaced by completion and Weil's bound in Lemma 4.2.

Friedlander and Iwaniec need a bilinear version, averaged over primes dividing the sequence, in order to push the half-dimensional sieve slightly past its trivial level. That gives the representability of $g(n)$ with the correct density $(\log X)^{-1/2}$ and without weights. Here we are content with a power of x , so the weight $r(N)$ suffices, and the loss of x^ε in Proposition 2.5 is harmless. Making their sieve argument uniform in d would gain only a power of $\log x$.

Remark 6.4 (The hyperbolic picture). Write $N = \frac{1}{2}\|\gamma\|^2$ with $\gamma = \begin{pmatrix} \alpha & \beta \\ \gamma' & \delta' \end{pmatrix} \in SL_2(\mathbb{Z})$, $d = \alpha^2 + \gamma'^2$ and $u = \alpha\beta + \gamma'\delta'$. Then $2dN = u^2 + d^2 + 1$ is the identity $\det \gamma = 1$. So the middles of triples correspond to the elements of the orbit $SL_2(\mathbb{Z})i$ whose norm $\frac{1}{2}\|\gamma\|^2$ lies in $\mathcal{S}$. The same equivalence between $n \pm 2 \in \mathcal{S}$ and the solvability of $ny = x^2 + y^2 + 1$ appears in recent work of Skafba [13].

The condition $d \leq x^\theta$ selects a thin region near the cusp, in which each fibre $\{\gamma : \gamma e_1 \text{ fixed}\}$ is one quadratic polynomial, and Hooley's method reaches level $\sqrt{x}$ there. For the full orbit, Friedlander and Iwaniec [5] prove level of distribution $x^{1/12}$ for $\|\gamma\|^2$ in progressions. A semi-linear sieve on the full orbit would need a level close to $x^{1/2}$.

7. NUMERICAL CHECKS

All identities used above (Lemmas 2.2, 2.3 and 4.1, and Step 1 of Proposition 4.3) were checked by computer on several hundred random instances. Table 1 compares $\sum_{w \geq 1, N_d(w) \leq x} r(N_d(w))$ with the prediction $4\sqrt{2} \mathfrak{L}_d \sqrt{x/d}$, which is the sharp-cutoff version of Proposition 5.1 ($F = \mathbf{1}_{[0,1]}$, $c_F = 2$), for $x = 10^{11}$. Here $\mathfrak{L}_d$ was computed from its Dirichlet series.

d	$\#\{w\}$	$\sum_w r(N_d(w))$	prediction	ratio	$8\mathfrak{L}_d$ vs. mean of r
17	54 233	215 068	216 134	0.995	3.985 / 3.966
41	34 921	114 120	115 669	0.987	3.312 / 3.268
73	26 171	70 504	71 827	0.982	2.745 / 2.694
89	23 702	55 968	55 623	1.006	2.347 / 2.361
97	22 704	125 520	124 339	1.010	5.477 / 5.529
113	21 035	61 024	61 163	0.998	2.908 / 2.901

TABLE 1. The weighted counts along $N_d(w)$ for $x = 10^{11}$ against the main term.

For $x = 10^8$ one finds $S(x) = 183\,099$ (counting $n = 1$), so $S(x)(\log x)^{3/2}/x \approx 0.145$. The family $\{N_d(w) \leq 10^8\}$ with d restricted to primes $d \equiv 1 \pmod{8}$, $d \leq D$, as in the proof, produces 3 687, 9 597 and 14 378 distinct triples for $D = 10^3$, 10^4 and $3 \cdot 10^4$. Allowing all $d \equiv 1 \pmod{8}$ gives 7 921, 22 284 and 34 593. The case $d = 1$ alone is the family $n = 8w^2 + 1$.

For $d = 17$, $h \in \{1, 3\}$ and M up to $1.28 \cdot 10^5$, the sums $\sum_\mu S_\mu(h)$ of Proposition 4.3 (with $\beta \equiv 1$) never exceeded about $0.4\sqrt{M}$ in absolute value.

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